AKHIL SEBASTIAN

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PROFILE SUMMARY

Final-year B. Tech Electrical and Electronics Engineering student skilled in Robotics and Project Development. Developed drone and robotic projects; proficient in Solidworks and Onshape; experience working with drone configuration software's. Demonstrates leadership, collaboration, and vested in research and development.

ACADEMIC QUALIFICATION

- Bachelor of Technology in **Electrical and Electronics Engineering** (CGPA: 8.16/10) 2022 - Present
  Mar Athanasius College of Engineering, Kothamangalam
- Senior Secondary (Percentage: 96.8%)
  Sainik School Kazhakootam, CBSE, Thiruvananthapuram 2019 - 2021
- Secondary Education (Percentage: 95.8%)
  Sainik School Kazhakootam, CBSE, Thiruvananthapuram 2014 - 2019

PROJECTS

8x8x8 LED Cube with Custom PCB | April 2024

- Concept: Designed and assembled an interactive 512-LED cube, integrating a custom PCB for matrix control and animation display.
- Tools: EasyEDA, Dot PCB, Arduino IDE, soldering station

3D Printed Drone Design and Assembly | IAM3D Competition | September 2024

- Concept: Designed, modelled and assembled a fully functional drone frame; integrated hardware and configured flight control systems.
- Tools: Betaflight, iNav, Mission Planner, SpeedyBee flight controller, 3D printing software, Solidworks
- Outcome: Developed a stable and responsive drone system; secured 2nd position in a national-level competition.

Drone Design Challenge 2025 | SAE | March 2025

- Concept: Designed, simulated and manufactured a functional box wing aircraft.
- Tools: Solidworks, XFLR5, Ansys CFD, Word, 3D Printing Software's

Design and Development of Autonomous Hybrid Robot for Multi-terrain Navigation | BTech. | June 2026

- Concept: Designed and developed an autonomous hybrid robot capable of navigating multiple terrains using sensor fusion, computer vision, and adaptive locomotion mechanisms.
- Skills Used: ROS2, autonomous navigation, embedded systems integration, motor control, sensor interfacing, image processing, path planning, hardware-software integration, PCB Designing
- Tools: Raspberry Pi 5, ROS2, Gazebo, OpenCV, LiDAR/ultrasonic sensors, Python, Solidworks, embedded controllers.]

KEY SKILLS

- **Technical:** PCB design, embedded programming, ROS2, Gazebo, MATLAB & Simulink, control systems, Betaflight, Mission Planner, 3D printing.
- **Soft Skills:** Documentation, project planning, system integration, troubleshooting, teamwork, collaboration.

INTERNSHIP

- Aircraft Design & Testing Trainee, **Feynman Aerospace**, Germany| January 2025 (30 Days)

Description: Participated in the design and performance evaluation of aircraft systems. Involved in hands-on testing, component analysis, and exposure to industry-standard protocols for aircraft validation.

- Robotics Project Trainee, **OpenDroids**, USA | January 2024 (30 days)

Description: Contributed to the development and refinement of a robotics project, assisting with hardware integration and embedded systems programming. Engaged in team-based troubleshooting and design review sessions.

Technology/Tools Used: CAD software, ROS2, Gazebo, embedded platforms.

- Testing and Validation Engineer, **AnoraLabs**, India | February 2026 (180 days)

Description: Worked on embedded systems testing and hardware validation for industrial electronic products. Assisted in debugging, protocol analysis, circuit-level testing, and validation of communication interfaces in embedded hardware systems.

EXPERIENCES

- **CHAIR, IEEE PES SBC MACE** | March 2024 – February 2025
  Led and coordinated technical events and workshops, enhancing leadership and organizational skills.
- **PROJECT LEAD, SAFETY DEVICES FOR WOMEN** | August 2024
  Managed the design and development of safety devices, improving team collaboration and product development skills.
- **TEAM LEAD, IAM3D Competition** | March 2025
  Directed drone design, assembly, and software setup, strengthening project management and embedded systems expertise.
- **CONTENT LEAD, IEEE PES Kerala Chapter** | April 2024 – March 2025
  Created and curated technical content, enhancing communication skills and technical knowledge dissemination.
- **CO-LEAD, DRONE DESIGN CHALLENGE** | October 2024
  Collaborated on drone project execution, fostering teamwork and problem-solving abilities.
- **LEAD, E YANTRA ROBOTICS COMPETITION** | August 2025 – Present
  Working with team members to develop autonomous crop monitoring drone using ros2, gazebo and image processing.
- **MATHWORKS MINIDRONE COMPETITION** | July 2025
  Created and designed SIMULINK Models and control systems for drone navigation using image processing.

LANGUAGES KNOWN

- English **(Fluent) |** Malayalam **(Native**) | Hindi **(Fluent) |** Japanese **(Basic)**